



Investigating the strategic choices of business ecosystem orchestrators: evidence from sports aggregation ecosystems

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Abstract

The business ecosystem literature offers valuable insights into various strategic variables such as pricing, governance, and network effects. However, few studies holistically analyze how orchestrators—the core firms that motivate and coordinate the activities of ecosystem partners—combine strategic choices to develop an ecosystem value proposition and compete against other ecosystems in the same domain. Therefore, we conduct an inductive multiple case study of German sport aggregation platforms and their ecosystems. We explore competing platform strategies by identifying relevant strategic variables, documenting ecosystem-specific dynamics and challenges, identifying patterns in the combinations of strategic choices, and tracking the development of strategic variables over time. Our findings contribute to the understanding of ecosystem strategies and support research on the intersection of strategic management and business ecosystems.

Keywords Business ecosystem strategy · Platform strategy · Ecosystem competition · Competitive dynamics · Competitive strategy · Strategic positioning · Strategic differentiation

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Abbreviations

PVP	Partner value proposition
CVP	Customer value proposition
BE	Business ecosystem
Aggregator	The orchestrator/focal firm of a sports aggregation ecosystem
Partner	Ecosystem participant, contributor, complementor

1 Introduction

Business ecosystems (BEs) have pervaded the global economy and affected billions of people (Evans and Schmalensee 2016) through different settings such as software development (Barlow et al. 2019; Boudreau 2010; Wareham et al. 2014; Zhou and Song 2018), app stores (Karhu et al. 2014; Rietveld et al. 2020), e-commerce (Curchod et al. 2020; Karle et al. 2020; Kwark et al. 2017; Zhu and Liu 2018), agriculture (van Dijk et al. 2023; Beishenaly and Dufays 2023), academia (Good et al. 2019; Miller and Acs 2017; Morris et al. 2017), healthcare (Lepore et al. 2023), and manufacturing (Rong et al. 2020, 2018; Kwak et al. 2018). Accordingly, BEs have attracted increasing research interest from different academic fields, such as engineering, information systems, sociology, law, and especially business (Kapoor 2018).

A BE can be understood as a conglomeration of collaborating actors creating a value proposition beyond each participant's capacity (Boudreau 2012; Jacobides et al. 2018; Kapoor 2018; Adner 2017). While Gawer (2021) emphasizes *platforms* as technical architecture that enables the creation of business ecosystems, Daymond et al. (2023) emphasize the existence of different ecosystem types that can be found in our empirical world and whose analysis constitutes different research streams (see also Cobben et al. 2022 and Martín-Peña et al. 2024). Business ecosystems extend well beyond the digital realm, and while many are built on digital platforms, our focus is not on technological architecture but rather on the strategic aspects of the business model. Network effects and related winner-takes-all (WTA) dynamics play a prominent role as sources of competitive advantage (Evans 2003; Rochet and Tirole 2003; Katz and Shapiro 1992; Cennamo 2018) and may induce strategic choices to unlock these effects (Eisenmann et al. 2011; Schilling 2002). However, network effects and WTA dynamics are not the only source of competitive advantage; strategies of differentiation can also play a role in competition between ecosystems (Cennamo and Santalo 2013; Huotari et al. 2017).

Strategic choices in a BE are made primarily by the ecosystem orchestrator who builds and nurtures the development and expansion of the ecosystem (Daymond et al. 2023). These strategic choices should not be analyzed in isolation¹ without consider-

¹ Various studies focus on specific strategic choices around pricing (e.g., Hagiu and Spulber 2013; Kwark et al. 2017; Karle et al. 2020), governance (e.g., Autio 2022; Boudreau 2010, 2012; Rietveld and Schilling 2021), subsidization (e.g., Amelio and Jullien 2012; Bakos and Halaburda 2020), and growth strategies (e.g., Bhargava et al. 2013; Staykova and Damsgaard 2015).

ing the implications on other elements of the overall ecosystem strategy (Krome and Pidun 2022). Strategy scholars (Porter 1996; Porter and Siggelkow 2008) emphasize that the focus should be on the combination of these closely interlinked decisions and that this combination is expressed in specific strategy configurations (Miller 1986; Mintzberg 1980), strategy groupings (e.g., Han et al. 2023; Thomas et al. 1999), or specific business models (e.g., Teece 2018). Surprisingly, while individual strategic choices are analyzed, few studies explicitly investigate the overall strategy combinations within ecosystems.

This study helps close this research gap by addressing two research questions: Which strategic variables do ecosystem orchestrators use to create competitive advantage? How do they combine strategic choices, and what is the rationale behind these strategy combinations? We apply existing knowledge of ecosystems to assess the dynamics and relationships between these choices.

We focus on Germany's sport aggregation market. In general, the sport industry represents a large and growing element of the global economy (Frisby 2005) and is becoming increasingly commercialized (Merkel et al. 2016; Hammerschmidt et al. 2022). This commercialization is in line with changing societal and economic demands and has caused the sport industry to evolve (Hoeber and Hoeber 2012). A growing body of research investigates sport entrepreneurship and how competitive pressures, the propensity for entrepreneurial activities, and inherent characteristics of the sport sector lead to business model innovation and corresponding shifts in management approaches (Hammerschmidt et al. 2022; Ratten 2010; Ratten and Jones 2020). We build on this research and focus on the emergent aggregation ecosystems in the sport sector (Hammerschmidt et al. 2023; Pellegrini et al. 2020). In this way, we also extend the ecosystem research realm by focusing on a tangible setting that is different from common use cases in the literature (e.g., software, app stores, and e-commerce).

This study presents five main findings. First, we empirically document the relevant strategic variables that allow for choices in the context of sport aggregation and extend the existing frameworks by introducing *partner benefits* and *network strategy* as essential strategic variables. Second, we empirically document the ecosystem-specific dynamics and show that different interests exist in the ecosystem, which the orchestrator must continuously balance. Third, we reveal that strategic variables are not independent but that patterns of choices emerge. Fourth, we empirically document that network effects and WTA dynamics are not the only sources of competitive advantage but strategies of differentiation can also be deployed. Finally, we document the development of strategies over time and show how the importance of individual strategic variables changes.

2 Theoretical background

2.1 Business ecosystems

In a business ecosystem, multiple actors collaborate for value creation and simultaneously pursue their economic self-interest by competing for value capture, as they

are bound by neither integrated hierarchies nor arm's-length market mechanisms (Rietveld et al. 2019). This setting in which actors cooperate and collaborate simultaneously has been termed “coopetition” (Brandenburger and Nalebuff 1997). Coopetition has been identified as a promising managerial approach in the sport sector (Hammerschmidt et al. 2020; Morris et al. 2007; Gast et al. 2014). While multiple synonyms for business ecosystems exist, including two-sided markets, digital and software platforms, and innovation and transaction ecosystems, we follow the logic of Thomas and Autio (2020) and understand the business ecosystem as a conglomeration of actors that create a clear value proposition targeted at a defined audience (Adner 2017). BEs have certain benefits, such as the ability to innovate through external partners (Pellizzoni et al. 2019) or realize production and consumption complementarities without vertical integration (Cennamo et al. 2020; Jacobides et al. 2018).

At this point, it should be clarified that references to ecosystems have grown steadily over the last few decades, as the concept has been used as an analytical lens across domains. It is therefore not surprising that multiple terms describe partially overlapping concepts (Daymond et al. 2023). While commonalities and differences across various ecosystem types have been reviewed (Thomas and Autio 2020), Daymond et al. (2023) highlight that especially the concepts of business, innovation, and platform ecosystems are closely related, with differences in terminology driven by the respective research focus. For our purpose, we acknowledge that business ecosystems and platforms are inextricably tied. While a BE describes the phenomenon of an ecosystem orchestrator managing a group of loosely connected complementors that collectively create a value proposition, platforms are often seen as the key infrastructure on which the business model is based (see Wegner et al. 2024 for a detailed review of the platform domain). As such, we have grounded our approach in the wider ecosystem literature which includes research with a strong focus on platforms (e.g., Dushnitsky et al. 2022; Cennamo 2018, 2021; Cennamo and Santalo 2013; McIntyre et al. 2021). For our research, we define the object of analysis as follows: we are investigating sports aggregation ecosystems as an example of business ecosystems in which the ecosystem orchestrator manages a central platform that enables complementors and users to connect.

2.2 Business ecosystem strategy

Successful strategy formulation in BE requires “ecosystem thinking” (Adner 2017) as competition has moved to the ecosystem level (Cennamo 2018). Strategies have become more complex, and forming an ecosystem strategy has been likened to playing “three-dimensional chess” (Eisenmann et al. 2006). Multiple groups of actors must be brought together (McIntyre et al. 2021) to share the benefits and risks of value-creation processes (Adner 2006). An ecosystem orchestrator must offer users an attractive customer value proposition (CVP) (Cennamo 2018; Cennamo and Santalo 2013; Halaburda et al. 2018; Mantena et al. 2010; Rietveld et al. 2019; Zhou and Song 2018). However, in addition to a CVP, orchestrators must attract and retain partners to create a focal offer (Adner 2017; Krome and Pidun 2022). An orchestrator can attract the best partners to realize a superior customer offer by making the ecosystem more attractive than the competing ecosystems (Cennamo 2018). Vari-

ous studies examine this partner value proposition (PVP) and the tools available to develop it (Hilbolling et al. 2020; Kazan et al. 2018; Rietveld and Schilling 2021; Rong et al. 2020).²

The ability of an ecosystem to attract partners and customers to participate is an essential driver of its success. More ecosystem participants imply more interaction opportunities (Eisenmann et al. 2006) and a higher ecosystem value (Cennamo and Santalo 2013). More participants also imply the creation of more complementary products within an ecosystem (Rochet and Tirole 2003). Indirect network effects set in when an increase in participants and offerings in one group creates value for other participant groups (Hagiu 2014).

The ongoing debate on what drives ecosystem success has long been dominated by network economists, who purport that network effects drive ecosystem values (Armstrong 2006). Cennamo (2021) differentiates this view through his framework by arguing that strategies aimed at creating competitive advantage through ecosystems are multidimensional. The first dimension, ecosystem size, is reflected in the user and complementor network. Corresponding strategies can be aimed at unlocking WTA dynamics (Katz and Shapiro 1992) which may cause markets to favor a dominant ecosystem (Cennamo 2018; Gawer and Cusumano 2008). Dushnitsky et al. (2022) motivate their study by proposing that orchestrators formulate strategies for ecosystems mainly to unlock network effects.

As the second dimension, Cennamo (2021) highlights *distinctiveness* of equal importance in creating competitive advantage. Strategic choices related to distinct positioning can include the breadth of and variety within the product portfolio (Karaer and Erhun 2015; Cennamo 2018), differentiation via a superior quality offering (Rietveld and Schilling 2021; Panico and Cennamo 2015, 2022) or simply the ability to tailor an offering to particular user groups (Huotari et al. 2017).

2.3 Strategic variables in the context of BEs

The existing literature investigates such strategic choices. Dushnitsky et al. (2022) summarize the discussion of the strategic variables in McIntyre and Srinivasan's (2017) study by aggregating the findings into six relevant key strategic variables that can be broadly assigned to either *pricing* or *non-pricing* (i.e., regulating ecosystem design mechanisms). *Pricing* variables regulate monetization through ecosystem fees. The literature identifies three variables: the subscription fee charged to participants who join the ecosystem, the transaction fee paid per usage, and the allocation of both fees among different participant groups. The third variable is often used to incentivize participation in the early phases of an ecosystem (Caillaud and Jullien 2003). *Non-pricing* variables include the control of access to the ecosystem, the scope of its offering, and the bundling of different functions. Table 1 provides a more detailed description of the strategic variables.

Given the range of strategic variables, orchestrators can mix strategic choices into various combinations. Therefore, an analysis of the overall ecosystem strategy should consider all these variables and their combinations in an integrated manner.

²For a contextual review, see Krome and Pidun (2022).

Table 1 Overview of key strategic variables as aggregated by Dushnitsky et al. (2022)

Strategic variable	Explanation
Subscription fee	A fee levied upon participants who want to join the ecosystem, i.e., the price paid for access. A low or non-existing fee can stimulate fast growth by attracting many participants but may attract low-quality offerings. Implementing a fee increases the burden of joining and thus the likelihood of quality contributions and provides income to the aggregator
Transaction fee	The fee is occurred by ecosystem participants every time they conduct a transaction through the ecosystem and can be set at an absolute amount or at a fraction of the transaction value
Fee allocation	The allocation indicates which participant group pays a particular fee. The literature sees the allocation consideration as an important tool in solving incentivization challenges such as the chicken-or-egg problem. The ecosystem can attract the more price sensitive participant group by allocating lower fees to it. Once this group is on board, network effects attract other participants such that transactions occur
Accessibility	Decision on the access to the ecosystem allow the orchestrator to control the number and kind of participants that can enter and transact within the ecosystem. The orchestrator faces the tradeoff between an open policy to attract a broad base of participants which may lead to strong growth but low-quality offerings and a restricted policy in which only selected participants are allowed to enter and contribute to the ecosystem
Scope of offering (inclusivity)	Inclusivity focuses on the scope of products/ services offered on the platform and thus relates directly to the ecosystem scope. Orchestrators can decide to focus on a particular product or market segment or to address a broad market segment by including a wide range of different products
Bundling	An ecosystem's positioning depends on the number of functions that are provided as part of the value proposition. Ecosystems can sharpen their identity by providing exclusively one function or bundle together different solutions attempting to reach economies of scope or carry-over to other industries

However, the extant ecosystem literature mainly examines individual tools for positioning, and little is known about the combination of strategic variables and their interactions (Krome and Pidun 2022; McIntyre and Srinivasan 2017). Our study aims to respond to the call for holistic studies on an ecosystem's overall strategy (Boudreau and Hagiu 2009; McIntyre and Srinivasan 2017).

3 Methodology

3.1 The sport aggregation ecosystem

Given the limited theory and sparse empirical research on BE strategies, we conduct a multiple case study guided by the theory-building intention highlighted by Eisenhardt and Graebner (2007). This method involves using one or more cases (Eisenhardt 1989) and is particularly suited to answer “how” and “why” questions in unexplored research areas (Edmondson and Mc Manus 2007).

We focus on Germany’s sport aggregation market, which provides an appropriate context for our research questions. Sport aggregation constitutes well-defined ecosystems in which value propositions can only be realized through the collaboration of independent economic actors. Moreover, the market has existed for several years; therefore, a set of strategic choices for various players can be observed over time.

We define the market specifically as a service offering stemming from the collaboration of three distinct actors: sports facilities, aggregators, and users of sports facilities. While aggregators build a network of partner sports facilities external users can buy access to this network through monthly subscription fees. In turn, the partners granting access to their facilities are financially compensated by the aggregator. The different aggregators active in this market thus compete for both partners becoming part of the aggregator network and users wishing to access it. We are looking specifically at the German market which functions as a conclusive unit of analysis (Titze and Gronau 2019).

The sport aggregation sector is particularly relevant for studying orchestrator strategies in business ecosystems for multiple reasons. It presents a vibrant market in which multiple players compete based on ostensibly diverging strategies (Titze and Gronau 2024). The product is an analog and thus differs in nature from more commonly researched software development ecosystems. While sports aggregators rely on digital platforms to enable their value propositions, products cannot be distributed without physical interaction. At the same time, the activity around business model innovation, strategic changes, and entrepreneurship is high (Jones et al. 2017; Hammerschmidt et al. 2021, 2023; Titze and Gronau 2019, 2024). We focus on the German sport aggregation market because it is relatively large, contains multiple competing players, and is dynamic, with high growth rates between 2021 and 2023 in terms of the number of engaged ecosystem partners (~25%), user subscriptions (~90%), and aggregator revenue (~70%) (Titze and Gronau 2019, 2024).

A sport aggregation ecosystem consists of three distinct types of participants: an orchestrator, independent physical sports facilities (partners), and individual users (customers) of sports facilities (Fig. 1). Note that in industry jargon, the term “aggregator” is usually used instead of “orchestrator”; we follow this convention and only use the term “orchestrator” again in the discussion and concluding sections when we draw general conclusions. An aggregator can generally be defined as a service provider who selects and compiles something for customers and offers it to them for purchase (Duden, cf. Titze and Gronau 2019).

The business model is simple. The aggregator builds a partner base of sports facilities that allow “outsiders” to use the facilities on a “per-check-in” basis in return for

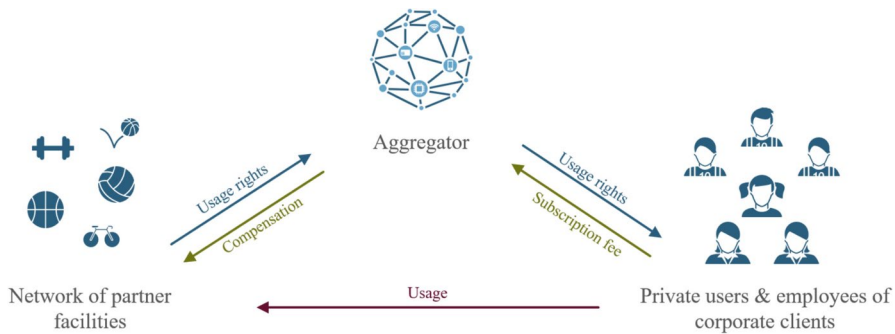


Fig. 1 Schematic overview of a sport aggregation ecosystem

the usage fee paid by the aggregator. However, the aggregator provides services to customers who pay a fixed monthly amount for flexible usage of the aggregator's partner network. These customers can be private individuals (B2C) or companies that offer the service as a subsidized benefit to their employees (B2B). This ecosystem solves problems for each participant group. When approaching large corporate clients, partners gain additional revenue from new customers without investing in marketing or sales. Users benefit from a flexible offer; they can use facilities irrespective of geographical location and take advantage of the breadth of sports offered within the network without committing to individual membership contracts.

We identified relevant ecosystems by screening industry databases and news sources for any firm active in Germany's sport aggregation. We verified our results using recent focused industry studies (Titze and Gronau 2019, 2024). Our sample comprises five firms operating in the German market. Table 3 (Appendix) provides an overview. We collected the respective data (see below) for firms and their ecosystems from their inception to early 2023.³

3.2 Data sources

We used the following data sources: (i) interviews with industry experts, formal bodies such as trade associations, and participants such as complementors; (ii) semi-structured interviews with employees of focal aggregator firms; (iii) informal follow-up interviews; and (iv) archival materials such as industry reports, company reports, news sources, and website data. We sourced historical data from the historical website information of firms stored on the Internet Archive Wayback Machine.

The semi-structured focal firm interviews comprised three main sections. The first section covers the respondent's background, role, and location within the focal firm. The second section attempts to understand the firm and its position within the competitive landscape, particularly the aggregator's strategy and its relationship with its competitive context. The third section addresses the relevant strategic variables and

³The German sport aggregation market comprises few competitors, most of which are in private hands. Due to this and the high degree of competition within the industry, confidentiality is a major factor. Thus, we are unable to disclose more information on the individual firms to protect their identity.

investigates the rationale behind the firm's specific strategic choices. If necessary, informal follow-up interviews, e-mails, and phone calls were used to fill in the gaps and verify the information. In addition to formal interviews, we used informal conversations with external participants to scrutinize the information received, improve our understanding of the overall industry, and obtain unbiased views when comparing different focal firms. The interview partners were identified via desktop research and selected using purposive sampling (Easterby-Smith et al. 2012), based on their roles in their companies, with a particular focus on either CEOs, sales staff facing the consumer side, or partner managers facing the partner side. An overview is provided in Table 4 (Appendix). Potential interview candidates were approached either directly via their professional e-mail addresses or via the professional network LinkedIn. Following similar approaches in comparable settings (e.g., Hammerschmidt et al. 2021), we deemed 17 interviews sufficient, given the relatively homogeneous population, and observed theoretical saturation (Saunders and Townsend 2018). Each interview lasted 45–60 min.

To ensure data validity, multiple respondents from each company were interviewed. A combination of internal and external respondents creates a better understanding than a single informant does. In addition, we used non-directive questioning techniques throughout our interviews to yield more accurate results. Finally, guaranteeing anonymity for all informants helped produce honest and concrete content, including the internal thinking of the focal firms.

Finally, we leveraged archival data in the form of news and industry articles. In addition to approximately 50 individual news articles, we relied on 11 industry studies conducted by consulting firms and industry associations between 2015 and 2021. Moreover, aggregators' websites were a rich source of information regarding their offerings and development over time, and we scanned approximately 1,250 archived aggregator websites using the Wayback Machine Internet Archive.

3.3 Derivation of strategic variables

We derived strategic variables relevant to our setting in three steps. First, we followed Dushnitsky et al. (2022) who had in turn relied on the literature review on platform strategies by McIntyre and Srinivasan (2017). We used the six key strategic variables identified (see Sect. 2) to define the option space. Second, we conducted exploratory interviews with aggregators and industry experts to assess the relevance of the six strategic choices and how they are operationalized in the sport aggregator setting. We also sought to explore additional variables that might be relevant in this specific context. Third, we interviewed the ecosystem partners to verify our insights. Interviewing both sides of the business relationship ensured data completeness.

We observed that the existing framework of key strategic variables by Dushnitsky et al. (2022) provides an excellent starting point for our analysis (see Sect. 2). However, our analysis of the archival material and the interviews led us to two observations. First, not all the provided variables are relevant to our context, as the non-pricing decision of function bundling does not apply to sport aggregation ecosystems focusing on providing a single core value proposition. Second, we identified two additional variables that are not part of the initial framework: *partner benefits*

(i.e., what an ecosystem can provide to attract and retain partners beyond monetary compensation) and *network strategy* (i.e., the decision on where and how to build a network of partners). We assume that these variables are not only relevant to our setting but also transferable to other ecosystem contexts. Therefore, the strategic variables used in our analysis are as follows:

1. The *channel* defines the accessibility of the ecosystem as the pathway through which the aggregator sells its service. The aggregator can target end users directly (B2C) or sell its services to corporate clients (B2B), who, in turn, transfer it exclusively to their employees.
2. The *customer offer* is the scope of the ecosystem in terms of the breadth, variety, and quality of services provided to platform users. In our setting, the customer offer constitutes the range of sports included in the network and the quality range of partners.
3. The *customer pricing* reflects how the customer is charged. The aggregator decides whether to differentiate its services through multiple price segments or offer only one uniform price. The industry generally relies on a monthly subscription fee model irrespective of price segment.
4. The *partner pricing* describes the monetary compensation scheme received by a partner for participating in an aggregator network. In our setting, partners receive compensation, from which the aggregator retains a fraction of the transaction fee whenever users check into a partner facility.
5. The *partner benefits* describe any contribution provided by the aggregator to increase its attractiveness to partners. It is an integral element of the PVP.
6. The *network strategy* reflects decisions regarding the geographic scope and density of a network in terms of geographic proximity among partners.

Figure 3 (Appendix) provides an overview of the transition and thought process.

3.4 Data analysis and theory building

We closely followed the process applied by Hannah and Eisenhardt (2018) in their study on the emerging solar panel ecosystems, which is particularly suitable for the nascent nature of our ecosystems. Accordingly, our study follows an exploratory approach given the sparse theoretical background of our particular research field (Hammerschmidt et al. 2021; Hannah and Eisenhardt 2018; West 2017). Thus, the thematic analysis is used as a method that offers an accessible yet theoretically flexible approach to organizing and describing a dataset. The thematic analysis highlights themes that are essential to understanding a given research topic (Fereday and Muir-Cochrane 2006) and yields rigorous results in complex surroundings (Nowell et al. 2017).

We began with a full transcription of all interviews using the *Trint* software. This step already involved the initial note-taking of relevant remarks, which we broadly attributed to the strategic variables that had been derived from the literature. We then structurally coded all interviews in correspondence with the identified variables and newly mentioned themes. If the data items were not related to the identified strate-

gic variables, new topics were created. Although individual respondents emphasized some themes more than others, we ensured that all data items were given the same attention. Next, we organized the developed codes into thematic (sub)categories while iteratively refining them to ensure internal homogeneity and external heterogeneity (Braun and Clarke 2006). After all topics had been reflected, we challenged the resulting overview to examine whether individual data points were sufficiently consistent to justify their appearance in the overall output. We iterated between transcribed data, mentioned interview topics, insights from the existing literature, and hypotheses built based on archival data. As a result, the outcomes were subsumed under six identified strategic variables: channel, customer offer, customer pricing, partner pricing, partner benefits, and network strategy.

A matrix view of the data allowed us to construct a holistic and comprehensive positioning profile for each focal firm. For each firm, we tracked the strategic variables from their inception and noted changes over time. After filling in missing details through follow-up e-mails, calls, or interviews, we confirmed the emerging patterns through the lens of our research questions.

After completing the within-case profile analysis, we examined the emergent patterns using a horizontal view across cases (Eisenhardt and Graebner 2007). We only highlighted information that emerged from multiple data sources or was emphasized by multiple interview partners. Using tables and visualizations of the spectrum of insights along with the strategic variables, we compared emerging theories and data to clarify concepts, develop categories, and refine abstraction levels. Thus, we followed an iterative process of refining our insights while building logical arguments (Eisenhardt 1989).

4 Findings

In this section, we present our findings for the six strategic variables. We highlight the observed differentiation of each variable and present the aggregators' rationale for a particular strategic choice. While this section describes these choices, their specific combinations are discussed in Sect. 5. Table 2 summarizes the results. Germany's sport aggregation market has only a few players, and its competitive intensity is high. To ensure confidentiality, we only provide fully anonymized information and refer to the firms as *Alpha* to *Epsilon*.

4.1 Variable 1: channel

The *channel* describes the route through which the aggregator sells its service. We observe two distinct channels. In B2C, the aggregator directly engages and contracts with individual end-users. In B2B, the aggregator engages with corporate clients and offers services to company employees. We observe two models: (i) B2B only and (ii) B2B and B2C.

Beta, *Gamma*, and *Delta* target only corporate clients (B2B) and rely on the active efforts of a sales team rather than on a strong brand image through marketing and advertising. By contrast, the B2C channel relies heavily on individual marketing.

Table 2 Overview strategic choices across focal firms

Aggregator	Strategic variables					
	Channels	Customer offer	Customer pricing	Partner pricing	Partner benefits	Network strategy
Alpha	B2B and B2C	Broad	Segmented	Fixed model	Omitted for confidentiality reasons	Urban focus, initially moving from large urban centers to next smaller ones, now extending into rural areas
Beta	B2B	Curated offering with opportunistic extension	Uniform	Variable model		Nation-wide, relatively less dense
Gamma	B2B	Curated offering	Uniform	Variable model		Nation-wide, relatively less dense
Delta	B2B	Broad	Segmented	Fixed model		Nation-wide, relatively less dense
Epsilon	B2B and B2C	Broad	Segmented	Variable model		Regional focus, relatively high density, no focus on cities

Alpha and *Epsilon* initially targeted B2C only but have eventually also entered the B2B channel, running both individual marketing campaigns directed at private end users and an active sales division that caters to corporate clients.

B2B clients are more concentrated than individual end users, making the channel easier to serve. One executive states, “*With B2C you have much more complexity and higher cancellation risks. And third, much, much more marketing expenses.*” B2B customers want their employees to utilize aggregators’ partner networks. Thus, sales efforts can be bundled into targeted negotiation processes, rather than addressing a mass audience. This results in a more “active” sales effort, but with the potential for stepwise growth as one *Beta* executive notes, “*B2B clearly means more investment in the beginning but at some point, it just is a cash cow.*” Furthermore, corporate clients invest in aggregator services for their employees, resulting in lower churn rates and less aggregator complexity. One executive states, “*The companies themselves have an interest that employees remain in the service and that is why you have extremely low churn rates. You have 2% compared to 20%.*”⁴

In the B2B channel, aggregators also benefit from subsidization because corporate clients financially support the use of aggregators’ products by their employees. This reduces the effective end-user price by as much as 80% compared to the B2C price when sold directly to the end-user.⁵ A lower effective price results in more subscribers and a higher participation rate. One executive claims, “*There you get many, many*

⁴ According to the respondent, 20% is the guidance number of annual member churn for an average fitness sports facility.

⁵ Interestingly, aggregators can charge a similar price in the B2B and B2C segment. The difference is from the subsidy that the corporate provides for its employees.

more subscribers than you would with a relatively high B2C price.” Another one adds, “You get above average participation rates, as you attract people who would normally never afford it. They would never pay 60 euros, but when it costs 25, they might do it.” Moreover, representatives from Beta, Gamma, and Delta argue that users who pay less also consume less, their consumption may lie below the price paid, as one executive explains, “You get people who tend to consume less as their expectations towards what they should consume are lower.” For example, we assume that the aggregator sells memberships to B2C subscribers at €50 per month. Meanwhile, B2B corporate clients partially subsidize the fee such that the end-user (the corporate client’s employee) only pays €20 and expects less usage. This attracts users whose consumption patterns are lower than the revenue they generate for the aggregator, thereby increasing profitability. One executive explains: “The more ‘couch potatoes’ you collect—they all pay the same—the more profitable it becomes.”⁶

This effect is also relevant for partners that join the ecosystem to generate additional revenue without converting existing (and potential) members into aggregator subscribers. Subsidized pricing is likely to attract new customers and bring in exactly those incremental users favored by the partners. Conversely, one executive highlights a higher risk of cannibalization for B2C as the channel is likely to attract customers who would pay a similar price for the usage of the partner’s facility anyway: “Exactly that you don’t have in B2C as each individual makes an optimization decision for himself based on price.”

Despite strong arguments for a B2B model, targeting B2C has several benefits, according to Alpha and Epsilon, including diverse marketing channels and cost-effective mass media. One executive notes, “B2C is easier because you can put an incredible amount of budget into marketing, [...] and people just sign up on their own.” An Alpha representative adds, “Our advantage is that we have a relatively high brand awareness through B2C.” This suggests that a strategic focus on branding and the quick ignition of network effects can aid B2B businesses. One executive explains, “Through this strong B2C brand that we have built, there is a lot of inbound B2B business.” In addition, the B2C channel is likely to attract users who have undergone personal optimization decisions, which makes them “active” users who need little activation. This benefits partners who are compensated based on the actual usage of their facility (see Sect. 4.4).

Over time, aggregators have increasingly focused on the B2B channel. Beta and Gamma have remained pure B2B players throughout. Delta terminates its B2C operations and focuses exclusively on B2B. Alpha and Epsilon, which started as B2C firms, have added B2B operations. Despite the attractiveness of B2B channels, Alpha and Epsilon continue to serve the B2C channel for brand and marketing benefits. In summary, aggregators consider channel choice as an important strategic variable.

⁶ Full quote for context: “For B2B you have to imagine a normal distribution and we have 5% of people who cost us real money; they are profit-negative, and it becomes better after that. And the more ‘couch potatoes’ you collect—they all pay the same—the more profitable it becomes. And for B2B you just get so many that the mix is right.”

4.2 Variable 2: customer offer

Customer offer describes the variety and quality of products and services available to the user. In our setting, it reflects the different sports and facilities that an aggregator includes in its partner network. For example, a narrow customer offer allows subscribers to access only a network of fitness studios, whereas a wide customer offer includes various sports in a single aggregator network. A wider offering improves the customer value proposition because optionality provides value. For the partner value proposition, increasing the breadth of the sport offering can also be beneficial, growing the partner network and enhancing its attractiveness without increasing direct competition between partners in any given sport type, because a broader offering is more likely to create complementary rather than substitutive services.

We observe three main approaches: (i) *Gamma* provides a small and concentrated offering; (ii) *Alpha*, *Delta*, and *Epsilon* provide a broad offering; and (iii) *Beta* creates a curated offering with opportunistic extensions.

Gamma's clear offering is a differentiating factor, as highlighted by one representative, "*The focus on particular segments is one of our differentiators as no aggregators execute it as diligently as we do.*" One executive explains how the clear communication of branding and value proposition creates a differentiated perception in the market, "*Alpha offers sports that we don't want to include, as the risk of injury is just too high [...], we say our core promise is health and not lifestyle.*" A clear brand focus, such as health benefits rather than leisure activities or a corresponding concise offering, can be communicated to customers more easily, thereby facilitating the sales process. One sales representative states, "*We manage to create a product that is easily understandable for people with limited interest in sport.*" Moreover, *Gamma* and *Beta* highlight that extending the breadth of the offering quickly impedes the ability to control quality within each type of sport. Another representative adds, "*I think what is important is the own quality standard. I don't think that it is about the quantity, but rather the quality.*" Therefore, a smaller offering may emphasize the curated development of quality standards over the rapid extension and growth of sport types.

Alpha, *Delta*, and *Epsilon* consider a broad range of sport types crucial for growth and customer attractiveness. As one *Alpha* executive notes, "*It is the breadth of the offering, and the density of the offering, that eventually determines the customer order.*" This view, shared by multiple aggregators, highlights that rapid user growth benefits from a broad offering across user interests. One executive explains, "*As we also do B2C, we try to much, much more emphasize lifestyle and we try to assume the users' view. And that leads to the fact that we have virtually all sports on offer that are somehow demanded.*"

Three different developments are observed over time. *Epsilon*, *Delta*, and *Alpha* expanded their customer offers, making breadth a core pillar of their value propositions. *Beta* started with a small range and has extended its offering slowly but consistently over the past few years to match different sport types, balancing curation with growth. Only *Gamma* remains focused on a highly curated portfolio, suggesting a strong focus on coherence and quality.

In summary, a broader *customer offer* helps create a compelling CVP that may support the faster growth of the user base. However, it may challenge sales personnel in communicating offerings and impede the maintenance of brand coherence and quality. Moreover, a tradeoff exists between addressing a larger market and focusing on a well-curated subset of the market. Thus, aggregators develop different strategies over time to address this tradeoff.⁷

4.3 Variable 3: customer pricing

Customer pricing refers to the manner in which aggregators charge the subscribers of a partner network. We identify two different approaches: (i) *Beta* and *Gamma* offer uniform pricing and (ii) *Alpha*, *Delta*, and *Epsilon* offer segmented pricing.

In a uniform pricing scheme, the aggregators offer a single product price to all subscribers. In contrast, in segmented pricing, users choose between packages at different price points corresponding to certain subsets of the aggregators' partner networks or services, defined as either access to certain partners or usage limitations.

Beta and *Gamma* use uniform pricing, arguing that it simplifies communication with customers and, thereby, the sales process. This simplicity benefits users by avoiding the need to check their demands for available offers and the partner network. A *Gamma* executive notes, "It needs to be as simple as possible for the end user. When I have different categories, I already discourage those who don't want to deal with that." In addition, a uniform approach may improve customer experience, as no restrictions apply. In contrast, and especially for B2B offers, the "creation of a two-tier society"—as phrased by one *Beta* sales representative—within one corporate customer can drastically reduce the offer's appeal. Finally, *Beta* and *Gamma* argue that a uniform price improves overall profitability. One *Beta* executive explains when reflecting on segmented pricing: "Through this supposed value difference you create a hierarchy and people who are not interested in sports [...] stay in the cheapest membership." Whereas users facing segmented prices can optimize their decision and choose memberships below a comparable uniform price, many users in a uniform scheme may subscribe to the service at a higher price, but may not fully use it (see Sect. 4.1).

Alpha, *Delta*, and *Epsilon* use a segmented pricing approach and highlight several benefits. First, different price points lower access barriers and attract potential customers, thereby fostering growth. Simultaneously, price segments allow aggregators to invest in more expensive partners to enhance the general network and brand. An *Alpha* executive highlights, "You try to restrict certain partners for certain price segments [...] as otherwise it simply would not be profitable for us."⁸ The argument

⁷This observed tradeoff is well-documented in the literature and reflects the discussion on whether an ecosystem invites partners freely, which facilitates growth and versatility, or whether partners are controlled for quality and ecosystem match, which hinders growth but improves user experience. For further reading, refer to Cennamo (2018), Zhou and Song (2018).

⁸For example, the highest price category user can access the whole partner network including premium partners. Premium partners are offered a higher payout for their facilities, which is compensated by the user's high monthly fee. The cheapest price category user can only access a part of the network—just those partners whose payout makes the user profitable for the aggregator.

proposes that providing price categories to users allows the aggregator to invest in a more differentiated quality range on the partner side while managing profitability risk. Segmented prices allow aggregators to control access to premium partners. This reasoning also applies to the partners' perspective. Premium partners may not want users from the aggregator's lowest price point to access their premium facilities, because their value proposition rests on providing a high-quality offering, both in terms of physical infrastructure and a uniform, controlled customer base. Finally, *Alpha*, *Delta*, and *Epsilon* propose that offering price categories makes them more attractive to customers. An *Alpha* sales executive explains, "*We have an S, M, L where you have the possibility—through different product prices—to provide each person access to a product and [...] the option to upgrade to a larger product.*"

We observe that the choice of pricing model is a tool for managing the breadth of the market segments addressed. *Beta* and *Gamma* address a narrower market segment through their uniform price in terms of the absolute number of potential users compared with *Alpha*, *Delta*, and *Epsilon*, which cover different price categories and have more scope to include a wider quality range in their partner networks.

In general, we observe few changes in customer pricing over time. *Beta* and *Gamma* have offered uniform pricing since their inception and reverted to it after experimenting with segmented packages. *Delta* and *Epsilon* have consistently provided five and three pricing segments, respectively. Only *Alpha* further differentiated from three to four segments in 2020.

4.4 Variable 4: partner pricing

Partner pricing is the monetary compensation received by a partner and is a major incentive component in building a partner network. One executive explains, "*The payment they're going to receive also will shift their interest between going to Gamma or Alpha.*" The partners are compensated by the aggregator each time a subscriber physically enters their facility.⁹ The aggregators retain a percentage of this compensation as a transaction fee. The compensation amount can be set through two different approaches: a fixed model, in which the amount paid per check-in remains the same irrespective of the number of check-ins, and a variable model, in which the compensation amount is adapted to the actual usage of subscribers.

Alpha and *Delta* use a fixed compensation model. The compensation amount is contractually agreed upon; however, aggregators also limit the total monthly payout generated by any single partner. As one *Delta* executive explains, "*For all fixed-price models it is usually capped [per user], the studios don't earn more than their own membership value.*"¹⁰ This model is simple to explain to partners and limits the aggregators' payout risk because aggregators can estimate the maximum amount paid

⁹Actual usage is when a user checks in at a partner location. This is done using the aggregator app and a physical identifier at the partner location.

¹⁰For clarification, the total payout to the partner is capped per individual user at the monthly value that the partner would charge the user for a regular membership on a long-term contract. For example, if a regular membership at the partner location costs €60, then, the partner is compensated up to €60 depending on the aggregator user's behavior. Beyond that, the aggregator user can continue to visit, but the partner is not compensated.

to partners. Because users can freely choose within the aggregator network and partners compete for subscriber loyalty, aggregators argue that a fixed model incentivizes partners to provide good services and attract users to their facilities. One *Delta* executive states, “Partners need to make sure [...] the user also goes frequently and then they are going to be able to almost receive their monthly membership.”

By contrast, *Beta*, *Gamma*, and *Epsilon* rely on a variable model in which check-in compensation fluctuates with subscriber usage. As one executive explains, aggregators place the collected subscriber revenue in a “pool” from which compensation is paid, “The whole money earned is put in one bucket, then the operating margin is subtracted, and then compensation is given from this bucket upon check-in.” With a fixed pool of money to be paid out, the amount paid per check-in decreases if users often check in to partner facilities, and vice versa. One *Gamma* representative highlights the collaborative aspect of this approach, “Collaborative compensation is very fair as it distributes the total company growth that we have equally to the partners. And that is substantially more sustainable for the partner.” While the model is less transparent and more complex to communicate, the aggregators purport that it reflects a collaborative approach to the ecosystem’s value proposition better. One *Beta* executive gives an example, “That is how all partners are participating in all our work. [...] we do not profit from users not going to the facilities.” While the fixed model focuses on quick growth, focal firms argue that the cooperative aspect of variable compensation creates a sustainable, partnership-focused, long-term growth trajectory.

Regardless of the motivation, the two approaches reflect different risk-sharing strategies between aggregators and partners. When the actual usage of subscribers fluctuates, aggregators can determine the extent to which the risk of losing revenue (revenue risk) and the corresponding impact on profits (profit-risk) are shared with the partners.

In the fixed model, aggregators receive a fixed monthly subscription fee from users; however, the payouts to partners depend on their actual usage. When subscriber usage fluctuates, aggregators have no revenue risk but have substantial profit risk. By contrast, partners face high revenue risk but lower profit risk. Their revenues are driven by actual usage but they equally incur lower variable costs as usage declines. As aggregators benefit from lower subscriber usage and partners benefit from higher usage, their interests are not aligned. However, the interests of users and partners are aligned because partners are incentivized to provide superior services to attract users to their facilities.

The dynamic changes in the variable model. Aggregators still have no revenue risk as subscribers still pay, irrespective of usage. However, aggregators have a lower profit risk because lower payouts per check-in compensate for the high number of check-ins. For partners, the variable payout reduces revenue risk because lower usage is partly compensated for by higher payouts per check-in. By contrast, partners tend to face increased profit risk. Higher usage increases variable (and opportunity) costs and reduces revenue per user, which puts pressure on margins. This change in the risk profile alters the incentive dynamics. In the fixed model, the aggregators benefit from low usage, whereas in the variable model, the interests of the aggregators and partners are better aligned. Conversely, the alignment between partners and

users decreases, as revenue smoothing according to usage decreases the incentive for partners to entice customers to return.

The aggregators' partner pricing choices remained consistent over time, indicating the importance of this variable in the ecosystem blueprint. Compensation structures, specifically partner pricing, can substantially impact (a) the attractiveness of the ecosystem for potential partners, (b) risk-sharing between aggregators and partners, and (c) incentives to improve the value proposition.

4.5 Variable 5: partner benefits

Partner benefits enhance value propositions and encourage participation in aggregator networks. Although financial compensation is important, aggregators offer more than monetary gains. *Partner benefits* include any stimuli offered by the aggregator to their (potential) partners. One executive states, *"That helps us to be perceived as a real cooperation by partners, not as a threat, really a partnership on eye-level."* Because payouts can be similar among different aggregators, non-contractual incentives can emphasize the collaborative partnership aspect and differentiate between ecosystems. Another executive provides an example, *"We have partners in our network that we continued to pay out during Covid, which goes well with our model."*

Aggregators provide various benefits to their partners. For example, they simplify the onboarding process and facilitate their partners' participation. Joining the network should *"not cause any additional effort,"* according to one manager. Another executive adds, *"Apart from physically placing materials at the site, nothing should remain in the hands of the partner."* Once onboard, aggregators strive to simplify operations, such as updating partner data in their systems. One executive notes, *"Data maintenance, it must be easy for the partner to maintain, digitally and with little effort."* Additionally, *Beta* and *Gamma* actively tailor marketing campaigns to specific partners and their locations to help partners increase facility usage. Aggregators also provide add-on services, such as perks. A *Delta* executive provides an example, *"We provide cash netness, so kind of secure their revenue,"* that is, a minimum compensation in case a threshold is not reached, or *"we have a private line in banks that is cheaper for Delta partners,"* or *"we provide payment matters like the payment machine for free."* Other aggregators use add-on services to increase their partners' attractiveness. One executive notes, *"We provide access to high-quality equipment [...] which is attractive to the partner's customers. Through our model, the partner can fully finance it without additional expense."*

However, potentially more interesting than short-term benefits are attempts to increase aggregator attractiveness in the long term. Representatives suggest that *"through the aggregator app, direct bookability into the partners' booking systems"* could streamline processes and optimize partner capacities. Furthermore, some aggregators envision a tailored credit system to actively smooth usage peaks for their partners. One executive explains, *"If I go at night during peak times, it costs me more credit points than if I come during lunch time."* Moreover, increasing integration can help aggregators access data that can be used to offer analytics services to partners to improve business operations or tailor offerings to customer needs. As one representative envisioned, *"When data in sufficient quality and quantity are there, I think, usage*

possibilities are endless, [...] then it could almost go into the direction of consulting of the partners."

Aggregators focusing on infrastructure differ from those focusing on marketing. *Beta* and *Gamma* provide active marketing aid for tangible cooperation results, whereas *Delta* focuses on providing IT infrastructure, thereby reducing cooperation. There is no clear dynamic visible as individual aggregators stick to the benefits they offer. Nevertheless, our interviews with both focal firms and partners indicate that the provision of benefits becomes more important as competition between aggregators for partners increases.

4.6 Variable 6: network strategy

The *network strategy* constitutes a major strategic variable in our setting due to the physical nature of the product. In a sport aggregation ecosystem, the *network strategy* defines the *geographic scope* and the *density* of the network in terms of the geographic proximity of the partners. The two variables are interdependent. Due to competition between ecosystems, no single aggregator can provide a nationwide dense network. Rather, aggregators either provide broad but relatively less dense network (e.g., *Beta* and *Gamma*) or a relatively dense network in fewer select locations such as cities (e.g., *Alpha*). Network growth can happen through both expanding geographically or densifying the network within a given region. The tradeoff influences network expansion strategies: a national 'loose' network can grow through densification, while a concentrated dense network can expand by entering new locations.

We observe three distinct patterns: (i) *Alpha* provides dense partner networks in urban agglomeration areas while expanding from one center to another; (ii) *Epsilon* focuses on a densely populated region across cities without aiming for national coverage; and (iii) *Beta*, *Gamma* and *Delta* provide relatively less dense nationwide networks without focus on highly urban areas.

The approaches of *Alpha* and *Epsilon* are similar as both aggregators address highly populated areas and can benefit from dense networks. While *Epsilon* focuses on a geographical region, *Alpha* is present in urban areas nationwide. Aggregators claim that populated areas are ideal for addressing the B2C segment (see Sect. 4.1). One executive explains: "*When I'm addressing the B2C customer directly [...] it makes sense to build agglomeration areas, because I can create a very large network in a very short time for users that I can reach with [television] advertising.*" This indicates that the envisioned growth pattern is also considered. Densely populated areas foster rapid growth, facilitating to build both partner networks and a customer base. However, rapid growth is linked to building dominance. For *Alpha*, one executive mentions: "*They clearly have such a WTA market share approach. They want to achieve such high market shares that they can reduce the check-in compensation*", implying that rapid growth partly aims for regional dominance. As creating dense networks is relatively easier in urban areas and network density is seen as a tool to become dominant, part of *Alpha's* rationale rests not only on growing quickly, but also on assuming the dominant position in a local market.

In contrast, *Gamma* and *Beta* focus on providing a geographically broad but relatively less dense network, arguing that customer type dictates network location. As

one executive notes, *"for those who do corporate sport [B2B], it is just much more important to see where the employees live"*, suggesting that B2B may not require a focus on urban areas but rather on the surroundings. While dense areas allow for rapid growth, *Gamma* and *Beta* focus on medium-sized cities, pursuing slower, yet more prudent growth. As one *Beta* executive reveals, *"what Alpha did quickly, almost too quickly as it created a huge cost surplus, that blew up in their face at some point, we try to do gradually."* In addition, one executive highlights: *"We choose organic, slow growth, rather not in urban areas, but in medium-sized cities [...] as the profitability is higher than in big cities."* Urban centers in Germany are limited; focusing on adjacent areas allows *Gamma* and *Beta* to build their networks while avoiding direct competition.

In summary, both *geographical scope* and *network density* can be used to foster network growth. While scope relates directly to which customer is addressed, network density is directly related to the CVP. Larger and denser networks improve the appeal of the value proposition, in line with the breadth of the sport offer, as discussed in Sect. 4.2. This would mean that the larger and denser the network, the more appealing it is to users. However, one manager contrasts this: *"We optimize the network such that we have exactly the right number of partners such that it motivates to choose one without being too many or too few."* Another one adds: *"We realize that too large an offer can overwhelm."* They suggest that aggregators face decreasing marginal user utility as the network grows and becomes more dense. Similarly, partners have ambivalent perspectives on network size and density. They benefit from being part of a large and dense network that is attractive to users. But a growing and especially increasingly dense network also increases competition within the ecosystem. As one executive explains, *"there is a point, fairly early, where the network gets afraid of more network growth, and which dislikes this growth as the pie is divided by more mouths."* This highlights partners' concern that increasing network density creates internal competition, eroding partners' willingness to collaborate. This effect is exacerbated in areas of market dominance, such that aggregators must balance network (density) growth carefully to satisfy both users and keep partners onboard (Fig. 2).

The decision on network strategy involves a tradeoff between geographic breadth and network density. Indefinite network growth seems an unsustainable strategy as it risks losing partners. One executive states, *"in the end, it is important in which amount to offer which sport to also make the partner happy that he can generate enough check-ins to stick with you."* Instead, the aggregator network must be carefully kept in balance as one executive insists on the need *"to get this ideal density where the studios [partners] are happy, where the users are happy, and where a sustainable ecosystem is created."* Aggregators must optimize network presence and density to create an appealing user offering while keeping partners content. Particularly for density-focused *Alpha* and *Epsilon*, geographical expansion without excessive network densification is necessary to grow further. As different aggregator networks expand, they begin to overlap in the limited German market, leading to eventual convergence to (and competition for) the same areas.

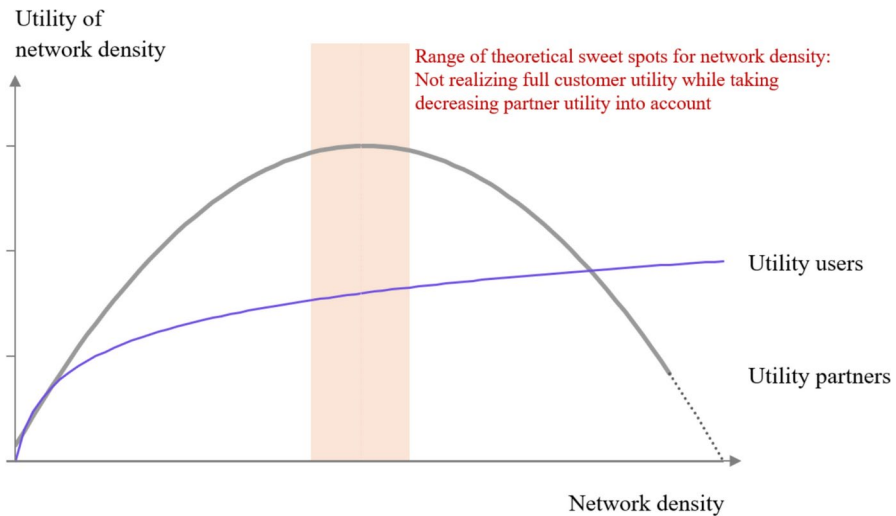


Fig. 2 Combined view on marginal customer and partner utility from network growth

5 Discussion

Our empirical analysis of sport aggregation ecosystems in Germany offers insights for scholars at the intersection of ecosystem and strategic management research and for the flourishing field of research on sport entrepreneurship (Hammerschmidt et al. 2021, 2022, 2023; Pellegrini et al. 2020). The previous section focused on the individual strategic choices made by sport aggregators in the German market. In this section, we analyze how aggregators combine different strategic choices, strategic rationale, and emerging patterns.

5.1 Relevant strategic variables

Our study empirically deduces six key strategic variables relevant to sport aggregation ecosystems: *channel*, *customer offer*, *customer pricing*, *partner pricing*, *partner benefits*, and *network strategy*. We leverage the results of Dushnitsky et al. (2022) who rely on McIntyre and Srinivasan (2017). We confirm the relevance of their three strategic decision variables related to pricing, as well as their non-pricing variables of *accessibility* and *scope of offering*. However, the relevance of *bundling* as a strategic variable that describes the range of functions offered by an ecosystem cannot be empirically confirmed. This may be temporary, as sport aggregator ecosystems are still relatively young and their functional offerings may expand in the future.

We identify two additional variables not included in the existing framework. First, *partner benefits* describe the benefits provided by aggregators in attracting and retaining partners in their ecosystem. As they are essential to the PVP (Cennamo 2021; Rietveld and Schilling 2021), we assume that they are relevant not only in our setting but also in other ecosystem contexts. Second, *network strategy* is essential because it builds the core infrastructure for delivering the value proposition. It comprises the

geographic scope and density of the partner network. Although the need to establish a physical network may drive the relevance of the network strategy in our setting, the variable may also apply to other ecosystems (Rysman 2009; Cennamo 2021).

In summary, our results confirm the findings of previous studies and extend the existing framework of strategic variables in business ecosystems. They encourage a more nuanced view of strategic variables that may be ecosystem-specific and context-dependent. To the best of our knowledge, research on how the ecosystem context influences the relevance of strategic variables is limited, and it provides fertile ground for future research.

5.2 Balancing interests within the ecosystem

An ecosystem is a loose conglomerate of independent economic actors who come together to create a value proposition (Reeves and Pidun 2022). To be successful, the orchestrator must motivate partners to join the ecosystem (McIntyre et al. 2021) and to keep them content and motivated to adhere to it (Adner 2017). Therefore, the orchestrator must develop not only a strong customer value proposition (CVP) but also an attractive partner value proposition (PVP).

Our study illustrates the potential impact of the partner pricing model (see Sect. 4.4) on the attractiveness of the ecosystem to potential partners, risk sharing between aggregators and partners, and incentives to grow and improve the ecosystem. This resonates with the existing literature on ecosystem governance models, which highlights how partners are integrated and managed in the ecosystem (Autio 2022; Boudreau 2010, 2012; Rietveld and Schilling 2021) or incentivized through subsidization (Amelio and Jullien 2012; Bakos and Halaburda 2020; Rochet and Tirole 2003) or general pricing schemes (Hagiu and Spulber 2013; Kwark et al. 2017; Karle et al. 2020). Our findings show how the aggregator needs to balance competing interests within the ecosystem and that certain tradeoffs arise. Compensating partners in the fixed model aligns the interests of users and partners because partners benefit when users return; partners have an incentive to provide superior services. However, aggregators benefit if users do not use partner facilities because of the lower compensation for partners. By contrast, in the variable compensation model, partners are less incentivized to provide good services because the compensation is adjusted to usage. The alignment between users and partners is reduced, whereas that between partners and the aggregator increases. Our example suggests that full interest alignment can never be achieved, and aggregators must carefully balance the interests of different participant groups.

The discussion on *network density* (see Sect. 4.6) further illustrates this point. We show that a partner network is a key driver of the customer value proposition. However, partners and users assess the network differently. Users prefer a larger and denser network because it creates optionality. However, a larger and denser network is not always beneficial for partners. Partners may face decreasing utility owing to increasing internal competition as the network grows denser. Our empirical setting confirms this phenomenon, known as the crowding-out dilemma (Cennamo 2021; Rysman 2009). This conflict arises from two competing forces. Although ecosystem participants derive value from a large pool of participants with whom they can inter-

act, they dislike overcrowding and competition within their respective participant groups. We suggest that there should be a theoretical sweet spot of network density (see Fig. 2) for every ecosystem. However, this is not easy to find in practice. Our interviews confirm that network strategy remains a major challenge for sport aggregators.

In summary, our findings highlight that ecosystem strategies involve tradeoffs that may require a constant balancing of interests among participant groups. This resonates with McIntyre et al.'s (2021) findings and underlines the fact that ecosystems must be viewed as dynamic constructs with a constant need for adjustment and rebalancing. Scott et al. (2022) come to a similar conclusion highlighting that the stability of ecosystems depends on the relations between actors which constantly over time. The process of not only establishing but maintaining balance over time has been largely unexplored in ecosystems but resurfaces as an essential topic that orchestrators must actively manage (Tsytsyna and Valminen 2024).

5.3 Ecosystem strategy combinations

Our findings show that orchestrators make different strategic choices for identified strategic variables, actively positioning their ecosystems in pursuit of competitive advantage. We observe certain patterns in how the strategic variables are combined, which is consistent with the notion of strategic groups (Thomas et al. 1999; Han et al. 2023) and configurational approaches, suggesting that although firms may differ in important characteristics, common patterns eventually limit their variability (Mintzberg 1979, 1980; Meyer et al. 1993; Short et al. 2008; Miller 1986).

For example, aggregators who choose the B2C channel (see Sect. 4.1) tend to be active in urban areas and create a concentrated and dense network. Densely populated areas provide a broad and scalable user base that can alleviate the challenge of convincing partners to join the network. Similarly, serving the B2C channel is facilitated by providing a broad customer offer (see Sect. 4.2). Unlike B2B, in which corporate clients partially make purchase decisions, B2C relies on individual end users buying into the aggregator's service. Offering a broad range of sports not only increases the value of the product for the user (flexibility) but also addresses a broader range of individual interests. Similarly, targeting B2C customers also helps in offering various price packages (see Sect. 4.3) as they span different price points, resulting in broader market coverage than a single price offer.

By contrast, the B2B channel (see Sect. 4.1) targets a different user type and benefits from a network that is rather broad in geographic coverage and less dense because the focus is on providing reliable access to the aggregator's service across locations rather than a lifestyle product. Similarly, B2B players prefer to offer a single uniform price rather than price segments to keep the product simple and easy for users to access.

We also find that aggregators who implement the variable partner pricing model (see Sect. 4.4) are likely to support their partners through active marketing campaigns. Aggregators can make this offer without regret. Partners benefit from aggregator support and an increase in visibility, while aggregators have no risk of higher costs from higher usage, owing to the variability in the compensation scheme.

Individual players leverage common strategic combinations. Ecosystem positioning results from different combinations of strategic choices because aggregators differentiate not only through individual divergences within strategic variables but also through diverging combinations of variables. Table 3 (Appendix) provides an overview of our sample.

The choices for *Beta* and *Gamma* are similar. Both ecosystems are active in the B2B segment and have broad networks with national coverage. This makes sense as their users are likely not concentrated in urban areas but rather dispersed at various corporate client locations across the country. In addition, both aggregators offer a mostly curated product portfolio, suggesting that the main focus is on the provision of an organic and accessible value proposition.

Similarly, *Alpha* and *Epsilon* show similarities in that both ecosystems serve the B2B and B2C segments while providing a broad customer offer and a segmented pricing approach. This combination suggests that rapid growth is a relevant objective. The broad customer offer and segmented pricing approach allow us to address a broader market, thereby facilitating the rapid growth of a strong user base. The network strategy shows similar logic. The focus is on populated areas, that is, large urban locations nationwide in the case of *Alpha* and a densely populated region in the case of *Epsilon*.

Delta diverges from these two clusters. Choosing B2B as the only channel is consistent with its mostly curated product portfolio and organic growth approach. However, its broad product offer and segmented pricing model seem to contradict the B2B logic. Given that *Delta* initially served the B2C channel, the divergence from the observed combinations of strategic variables may be due to the transition from one model to another. Adherence to the segmented pricing model, which is uncommon in the B2B segment, can also be understood as an experiment on whether users appreciate the increased flexibility.

In summary, our study shows that orchestrators use combinations of strategic variables to position their ecosystems actively in pursuit of competitive advantage, underlining the argument that strategies should be viewed holistically (Porter 1996; Porter and Siggelkow 2008). This is consistent with Dushnitsky et al.'s (2022) finding that combining choices with strategic variables can be used to create value. However, we advance the results of Dushnitsky et al. (2022) by finding evidence that the ecosystem size is not the only rationale. Some aggregators in our sample deliberately abstain from creating large networks to focus on distinct positioning. This challenges the views of many network economists (Armstrong 2006; Rochet and Tirole 2003) who consider network effects as the main driver of success and the pursuit of WTA effects as a superior strategy in ecosystem competition (Katz and Shapiro 1992; Gawer and Cusumano 2008). However, this finding is consistent with that of Cennamo (2021), which highlights *distinctiveness* as a second potential dimension for creating competitive advantage in ecosystem competition.

5.4 Developments over time

For the choice of *channel*, we observe a general trend of moving away from the B2C-only model to either a pure focus on B2B (*Beta*, *Gamma*, and *Delta*) or add-

ing the B2B channel to the existing B2C approach. (*Alpha* and *Epsilon*). As such, the range within the variable decreases over time; however, the two major choices remain differentiators.

Customer offer as a strategic variable shows a clear trend toward convergence as most players have extended the breadth and variety of their offerings. *Alpha*, *Epsilon*, and *Delta* actively communicate this as part of their value propositions. *Beta* has long stuck to offering a highly curated portfolio but has recently started to expand it opportunistically. *Gamma* is the last player to focus on a small curated offering.¹¹

Customer pricing has become increasingly stable over time. *Beta* and *Gamma* have provided uniform prices since their inception, with only a few attempts to move to price segmentation. All other players started with different price segments and only increased the number of segments for differentiation purposes (i.e., *Alpha* went from three to four price options).

Similarly, *partner pricing* has not changed over time. Although the players apply different models, they still operate under the same initial model. Given the importance of the compensation model for ecosystem governance, risk sharing, cooperation, and trust among individual participant groups, this variable may not be easy to change.

Finally, *network strategy* has exhibited the largest development over time. While the ecosystems started with different strategies (e.g., *Alpha* vs. *Gamma/Beta*), we observe an increasing convergence. This may also be due to the geographic limitations of the German market and the initial overlap of competing networks. Aggregators focusing on dense urban centers (e.g., *Alpha*) have expanded beyond large cities once network density has reached a point where partners have started to show dissatisfaction because of increased within-ecosystem competition. This development is exacerbated by the inclusion of the B2B channel, which benefits from non-urban areas. *Beta* and *Gamma*, which started with a nationwide and relatively less-dense network, have built a strong and geographically broad network. Therefore, they can extend their presence to urban areas to make themselves more attractive.

In summary, we see both convergence and divergence within the strategic variables. The *customer offer* and *network strategy* show convergence by including more sports in the value proposition and increasing the overlap of physical partner networks. Both *customer pricing* and *partner pricing* show little change over time, and strategic choices can act as clear differentiators. Similarly, the decision to serve the B2C channel differentiates two aggregators from the others. Based on these trends within the variables, different strategy combinations form over time (see Sect. 5.3), to which the aggregators *Alpha/Epsilon*, *Gamma/Beta*, and *Delta* can be attributed. As the market is still relatively young, we expect further convergence within certain variables and an increasing need to differentiate through the remaining variables.

¹¹For a discussion on the tradeoff between curated and broad product offerings, please see Cennamo (2018), Karaer and Erhun (2015), Rietveld and Schilling (2021).

6 Conclusion

6.1 General conclusions and insights

We empirically examine German sport aggregator ecosystems and document several key findings. First, we identify the relevant strategic variables in the sport aggregation context and extend existing frameworks by establishing *partner benefits* and *network strategy* as additional strategic variables that are important for aggregation platforms, a dynamic sector in the emerging field of sport entrepreneurship (e.g., Hammerschmidt et al. 2021, 2022, 2023; Pellegrini et al. 2020). We highlight that no generalizable set of strategic variables can be applied to any ecosystem context. Instead, an ecosystem strategy may rely on ecosystem-specific variables that are essential for strategy formulation. Similarly, not all variables in existing frameworks apply to every setting. In our context, we demonstrate that *bundling* is not (yet) a relevant strategic choice. Scholars should not only rely on existing frameworks but also critically explore how the proposed sets of strategic variables should be adapted for a specific context.

Second, we document the ecosystem-specific dynamics and show that establishing a strong PVP is as important as having a superior CVP. Simultaneously, we reveal that orchestrators must continuously balance the different interests of their ecosystems. The *network strategy* discussion shows that building a strong CVP may impede the PVP because increasing the network density beyond a certain point benefits users but is not favored by partners. Similarly, the *partner pricing* example shows how an orchestrator's choice of strategic variables can change the alignment of interests within an ecosystem. One model aligns the interests of partners and users, whereas the other aligns the interests of partners and aggregators. Therefore, full alignment may not be achieved. To date, this dynamic view has not been covered in the ecosystem literature. Partners, as an additional unknown variable for the orchestrator, and the need to continuously balance diverging interests within the ecosystem suggest that strategies in the ecosystem context tend to emerge rather than be deliberate (Mintzberg 1987).

Third, we observe that the strategic variables are not independent, and patterns of combinations of choices emerge. For example, we show that firms targeting B2C customers (private users) may benefit from broad product offers and segmented pricing. Conversely, firms focusing on B2B customers may benefit from curated product offers and uniform pricing. This confirms the relevance of configurational approaches (Miller 1986; Mintzberg 1979, 1980), strategy groupings (Thomas et al. 1999; Han et al. 2023), and specific business models (Teece 2018) in an ecosystem context.

Fourth, orchestrators in the sport aggregation context leverage consistent combinations of strategic choices to position their ecosystems in the pursuit of competitive advantage. They do not exclusively pursue strategies to reach a dominant market status and unlock WTA dynamics but rather create differentiated and distinct positioning. This strongly confirms Cennamo's (2021) framework by highlighting the tradeoff between pursuing a strategy aimed at size and the creation of value through distinct positioning. Furthermore, this insight challenges the widely held view that network effects are the main value driver (Armstrong 2006; Katz and Shapiro 1992) and, thus,

the main source of competitive advantage in ecosystem competition (Eisenmann et al. 2011; Evans 2003; Rochet and Tirole 2003).

Our empirical investigation of multiple competing ecosystems and their strategic development over time contributes to the literature on the intersection of business ecosystems and strategic management. We do not develop or test theoretical hypotheses as this is an inductive study. Rather, we validate and extend the existing research. Specifically, we extend Dushnitsky et al. (2022) and supplement empirical studies that focus on the strategic decisions of selected platforms (Boudreau and Hagiu 2009; Li and Pénard 2012; Schilling 2002) or use data from several platforms to test hypotheses regarding certain individual strategic choices (Cennamo and Santalo 2013; Claussen et al. 2013; Zhu and Iansiti 2012).

Finally, our study contributes to the emerging and promising field of sport entrepreneurship by introducing orchestrators and partners as new actors and units of analysis into the research context. While existing literature has focused on traditional actors such as football clubs (e.g., Hammerschmidt et al. 2021), the sport sector is arguably equally affected by ecosystems as innovative business models.

6.2 Limitations and future research

This study has some limitations that offer opportunities for future research. First, our study is geographically limited because it analyzed ecosystem players in Germany. While this was intentional due to our clear definition of the market, it may limit the generalizability of the results. Other markets suggest that geographical context can make a difference. For example, the business models in the US sports aggregation market differ from those in the German market. In contrast to the subscription flat rate granting users access to the partner network in Germany, players in the US tend to rely on a pre-paid credit model (fitness MANAGEMENT 2019), suggesting differences in customer or partner preferences. Meanwhile, the Polish sports aggregation market operates under the same value proposition seen in Germany, yet its market dynamics have developed very differently. Instead of competition among multiple actors, one aggregator came to dominate the market and even entered the domain of sports facilities by exerting pressure on partner compensation and acquiring financially challenged facilities for direct competition (fitness MANAGEMENT 2019).

Second, our study does not include additional contextual factors (e.g., regulations, marketing, technology) that may influence the observed mix of strategic choices. Similarly, internal firm characteristics such as management capabilities or general organizational maturity could explain some of the observed strategic choices.

Third, our study is subject to general limitations that researchers have pointed out for the multiple-case study methodology. While case studies can effectively capture contemporary phenomena (Käss et al. 2024), some researchers argue that the method lacks rigor, particularly regarding the validity of findings (Dubé and Paré 2003; Lee and Hubona 2009). And while Hollweck (2015) shows that structured approaches to case study research can ensure sufficient rigor, controlling researcher bias and replicating interviews in the same manner remain challenging limitations.

Naturally, our research provides a foundation for future studies. First, our findings should be tested and validated in different ecosystem contexts. Our results show that

strategic variables may differ depending on each ecosystem and its value proposition (see Sect. 5.1). Therefore, the unique characteristics of an ecosystem must be considered when identifying not only relevant strategic variables but also their combinations. Future research could test whether differences in outcomes are evident to infer whether the results are generalizable, i.e., whether strategic variables are always used for differentiation and whether patterns emerge in every setting. We propose the aforementioned “structured” approach to test this. On the one hand, researchers could examine an ecosystem with similar characteristics to our setting but with a different product—for example, a digital infrastructure used to connect supply and demand where both sides are only loosely connected as is typical for a BE. A good example is the ride-hailing industry, where the ecosystem orchestrator must attract drivers and users to its platform. The results would reveal whether the results are influenced by industry sector. On the other hand, researchers could test whether the results vary across ecosystems that have different characteristics. An example is the purely digital realm of mobile operating systems, where physical infrastructure—such as sports facilities in our case or cars in ride-hailing—does not matter for product delivery. Similarly, the degree of product complementarity may be an additional differentiator. Whereas our setting shows low levels of complementarity—sports facilities also function without aggregator or other facilities in the ecosystem—other settings, such as video games and consoles, show high complementarity, i.e., both are required for the product to work. It would be insightful to explore whether, given these characteristics, strategic variables are consistently used for differentiation and whether patterns comparable to our results emerge.

Second, future research should place more emphasis on how strategic choices in ecosystems affect performance. While Porter (1996) defined successful strategy mainly through competitive differentiation, the literature increasingly views strategic success as something comparable and measurable through performance (Barney et al. 2023). Therefore, future research must develop appropriate performance measures, which depends on data access. In our setting—young and privately held firms—data constraints prevented the development and assessment of suitable performance measures. However, this assessment is generally relevant, requiring creative approaches that define performance measures and link them to strategic choices. While Fernández-Portillo et al. (2024) highlight a positive correlation between digital ecosystem engagement and a loose definition of firm performance, Dushnitsky et al. (2022) provide a strong example of how especially different combinations of strategic choices may be evaluated in terms of performance. Overall, future research may need to devise performance measures for the ecosystem context more broadly and from a conceptual standpoint. Some research seems to equate platform success with the platform size or the number of ecosystem participants (e.g., Cennamo 2018, 2021). Future research could explore whether ecosystem performance can truly be reduced to goals such as market dominance, or whether more traditional success factors such as profitability or shareholder value should be considered.

Third, with an objective measure of performance, future research could further examine additional success factors. Our results show that certain combinations of strategic variables emerge, suggesting that there is no single best solution for market positioning (see Sect. 5.3). This is in line with configurational theories purporting that different sets of conditional variables can explain desired outcomes (Dellermann and Reck 2017). Future research could apply these configurational approaches to ana-

lyze strategic variables (e.g., Zhao et al. 2017) and understand which combinations lead to successful outcomes while considering respective contextual factors (Miller 2018). Greckhamer et al. (2018) argue that qualitative comparative approaches such as fsQCA¹² are especially suitable for moving beyond describing configurations (e.g., Greenwood and Hinings 1993; Miles and Snow 1978). By applying fsQCA, researchers could measure different sets of strategic choices (CVP, PVP, network strategy) against ecosystem performance. Beyond internal choice parameters, future research could extend the analysis by including contextual factors such as market development or maturity, the degree of regulatory requirements, or complementor-specific characteristics. This would not only clarify why particular sets of strategic variables are chosen, but also reveal which other factors may influence ecosystem success.

Fourth, future research should emphasize how strategic choices evolve over time, which is interesting for several reasons. On the one hand, a time-series analysis could investigate whether convergence occurs across strategic choices or whether particular strategy combinations reduce variability within the range of strategic variables. This would again support configurational approaches arguing that although firms may differ in important characteristics, common patterns eventually limit variability (Meyer et al. 1993; Miller 1986; Mintzberg 1979; Short et al. 2008). Our research adds a temporal perspective (see Sect. 5.4), showing that some variables converge. Future research could thus address whether there is convergence towards one superior model or whether the push for competitive differentiation persists. On the other hand, our analysis highlights that firms develop their business models over time. Utterback and Abernathy (1975) contextualize these changes by the age and developmental stage of each respective industry, arguing that young industries are highly innovative in the product dimension, yet their processes are largely undefined and emerging. As an industry matures, product innovation decreases and process innovation becomes more important. Both the sports aggregation industry and its ecosystems as an organizational form are relatively young. Future research could investigate to what extent ecosystems follow a similar development pattern, shifting from product innovation toward process innovation. In particular, studies could analyze whether the basis of differentiation changes, i.e., whether ecosystems focus on product differentiation in an industry's early stages and then shift to process differentiation as the industry matures. Furthermore, future research could compare ecosystem development patterns with "traditional" industrial evolution patterns, thereby shedding light on creativity and innovation as topics that have recently attracted significant interest in the sport entrepreneurship literature (Hammerschmidt et al. 2023).

6.3 Managerial implications

Our study provides managerial insights that can be generalized to practitioners in other ecosystem contexts. First, the study is one of the few to holistically analyze platforms' strategic choices and the related dynamics and tradeoffs within the ecosystem. Orchestrators combine their strategic choices to create distinct positioning profiles. This suggests that there is more than one way to succeed in ecosystem com-

¹² Fuzzy-set qualitative comparative analysis.

petition; it is not just about building the largest network most quickly. When defining and addressing a matching market segment with a relevant and differentiated offering is key, orchestrators can succeed by positioning themselves according to the diverse preferences of distinct customer groups (Cennamo and Santalo 2013; Hossain et al. 2011; Huotari et al. 2017). Ecosystem managers should not simply copy existing business models but rather develop proprietary strategies that diverge from the strategic combinations undertaken by competing ecosystems.

Second, our results highlight the need to balance diverging interests within an ecosystem, which should be reflected in the strategy-making process. Our findings regarding partner pricing models and network strategy (see Sect. 4.6) provide good examples of how interests may diverge. In addition to a superior customer value proposition, orchestrators must develop an attractive partner value proposition to attract and retain partners and motivate them to be active in the ecosystem. In summary, our study encourages managers to implement a holistic approach to ecosystem strategy to better align individual elements and coordinate participants from distinct groups.

Appendix

See Fig. 3 and Tables 3, 4.

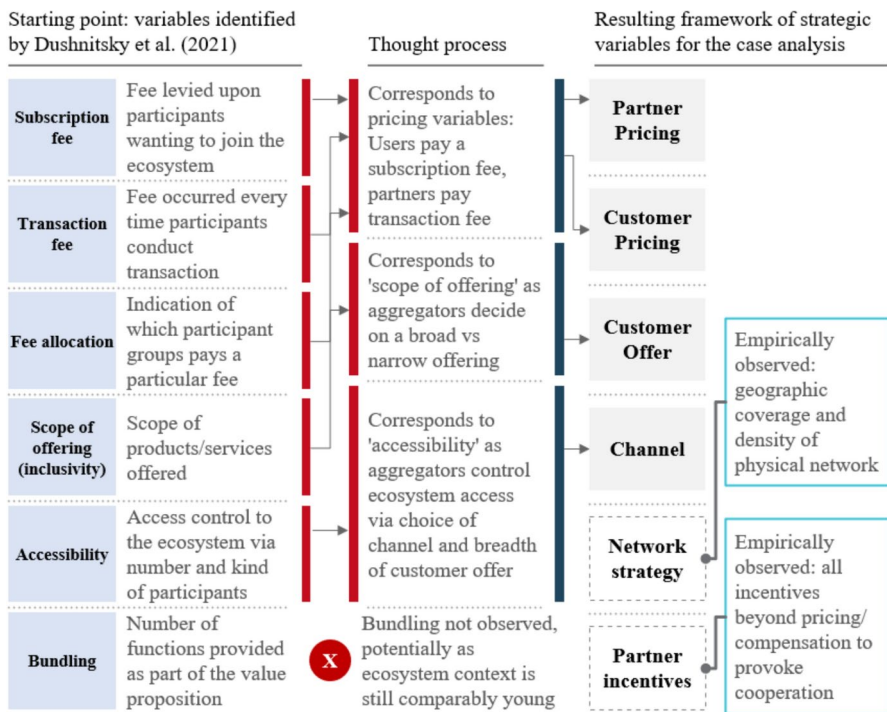


Fig. 3 Schematic overview of strategic variable transition

Table 3 Overview of the cases

Company	Synonym	Size	Time in existence	Roles interviewed	German market only
1	Alpha	Large	> 10 years	B2B/B2B Sales	No
2	Beta	Large	> 10 years	CEO Corporate development	Yes
3	Gamma	Large	> 10 years	Sales Partner management	No
4	Delta	Medium	< 10 years	Client success strategy	No
5	Epsilon	Small	< 10 years	CEO	Yes

Table 4 Overview of interview partners

Interview	Affiliation/aggregator	Role
1	Gamma	Senior manager (sales)
2	Gamma	Senior manager (sales)
3	Gamma	Senior manager (sales)
4	Leading industry association	Industry expert
5	Partner organisation (small fitness chain)	CEO
6	Beta	CEO
7	Leading industry association	CEO
8	Gamma	Head of partner management
9	Fitness management	CEO
10	Alpha	Head of B2B growth
11	Beta	Senior manager (sales)
12	Alpha	Head of B2B sales
13	Beta	Corporate development manager
14	Gamma	Head of partner Mgmt
15	Delta	Global client success strategy senior manager
16	Leading industry association	Industry expert, focus corporate wellbeing structures
17	Gympool	CEO

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article, no support in financial or non-financial matter was received.

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